QQ-W-345b
Amendment-2
July 2, 1968
SUPERSEDING
Interim Amendment-1 (GSA-FSS)
November 7, 1966

FEDERAL SPECIFICATION

WIRE, ELECTRICAL, STEEL, COPPER-COVERED

This amendment, which forms a part of Federal Specification QQ-W-345b, dated November 7, 1966, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

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Between paragraph 3.2.1 and 3.2.2 insert the following:

3.2.1.1 Tensile strength of joints. The wire shall be free from joints made in the finished wire. Joints may be made in the wire prior to the final drawing operation. These joints shall be the scarf type, brazed with silver-alloy solder. The solder shall completely cover the scarfed faces and edges of the joints. The joints shall be smooth, uniform, and free from lumps of solder. The joint shall have a breaking strength of not less than 85 percent of the adjacent wire at least one foot from the joint. The section of the wire containing a joint shall be capable of being wrapped around three times its own diameter without rupture of the brazing material, the copper coating, or the steel core of the wire.

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Section 5: Delete in its entirety and substitute the following:

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specifed (6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging.

5.1.1.1.1 Coils (nominal wire diameters 0.204 through 0.080). Wire sizes furnished with a nominal wire diameter of 0.204 through 0.080 shall be uniformly coiled in coil weights as specified in table III. Each coil shall be secured at 3 places with sisal twine having a breaking strength of 215 pounds, evenly spaced around the coil. Each coil shall have a peripheral wrap of paper conforming to PPP-B-1055, class B-2, wrapped around the outer circumference of the coil. Two spiral wraps of the same class of paper shall be wrapped over the peripheral wrap with a continuous length of corrosion resistant binding wire not less than 0.032 inches in diameter on each turn of the wraps. The spiral wraps of paper shall be edge folded and each turn shall overlap the preceding turn by 1-1/2 inches. The end of the wrap and binding wire shall be securely tied with corrosion resistant wire not less than 0.032 inches in diameter.

5.1.1.1.2 Machine-straightened wire. Machine-straightened wire shall be furnished in standard weight coils as specified (see 6.1). The minimum inside diameter of each coil shall be not less than 30 inches.

5.1.1.1.3 Spools or reels (nominal wire diameter 0.079 through 0.020. Wire sizes furnished with a nominal wire diameter of 0.079 through 0.020 shall be fully and evenly wound on returnable or nonreturnable wood or metal spools or reels in spool or reel weights as specified in table IV. The wire furnished on the spools or reels shall be wound to obtain uniformity, compactness and nesting of successive turns and layers. The end of the wire shall be secured to prevent unwinding. The spools or reels shall be designed with strong flanges of sufficient diameter to protect the wire during shipment and storage. The wire wound on the spool or reel shall be covered with a full

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wrap of material conforming to PPP-B-1055, class B-2, overwrapped with a full layer of kraft lined chipboard with a minimum thickness of 0.080. The wraps shall extend between the flanges and be secured with flat metal strapping. A full wrap of wired wood slats 1 inch by 1/4 inch shall be provided over the chipboard and secured with a minimum of two flat steel straps.

5.1.2 Level B.

5.1.2.1 Unit packaging.

5.1.2.1.1 Coils (nominal wire diameters 0.204 through 0.080). Wire sizes furnished with a nominal wire diameter of 0.204 through 0.080 shall be uniformly coiled in coil weights as specified in table III. Each coil shall be secured at three places with sisal twine having a breaking strength of 215 pounds evenly spaced around the coil. Each coil shall have a peripheral wrap of paper conforming to PPP-B-1055, class B-2, wrapped around the outer circumference of the coil. A spiral wrap of the same class of paper shall be wrapped over the peripheral wrap with a continuous length of corrosion resistant binding wire not less than 0.032 inches in diameter on each turn of the wrap. The spiral wrap of paper shall be edge folded and each turn shall overlap the preceding turn by 1-1/2 inches. The end of the wrap and binding wire shall be securely tied with corrosion resistant wire not less than 0.032 inches in diameter.

5.1.2.1.2 Machine-straightened wire. Machine-straightened wire shall be furnished in standard weight coils as specifed (see 6.1). The minimum inside diameter of each coil shall be not less than 30 inches.

5.1.2.1.3 Spools or reels (nominal wire diameter 0.079 through 0.020). Wire sizes furnished with a nominal wire diameter of 0.079 through 0.020 shall be fully and evenly wound on returnable or non-returnable wood or metal spools or reels in spool or reel weights as specified in table IV. The wire furnished on the spools or reels shall be wound to obtain uniformity, compactness and nesting of successive turns and layers. The end of the wire shall be secured to prevent unwinding. The spools or reels shall be designed with strong flanges of sufficient diameter to protect the wire during shipment and storage. The wire wound on the spool or reel shall be covered with a full wrap of material conforming to PPP-B-1055, class B-2 overwrapped with a full layer of kraft lined chipboard with a minimum thickness of 0.080 inches. The wraps shall extend between the flanges and be secured with a minimum of two flat steel straps.

TABLE III.

Nominal wire diameter	Coil weight		Dimension of coils
	Maximum average	Minimum average	Inside diameter minimum
Inch	Pounds	Pounds	Inches
0.204 - 0.128	225	175	18
.127 - .100	225	125	18
.099 - .080	225	50	18

TABLE IV.

Nominal wire diameter	Spool or reel weights	
	Maximum average	Minimum average
Inch	Pounds	Pounds
0.079 - 0.060	200	100*
.059 - .040	200	100*
.039 - .020	200	100*

*At least 90 percent of the spools or reels in any one shipment shall have weights not less than 100 pounds and the remaining percent shall have weights not less than 60 pounds

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5.1.3 Level C. The wire furnished in coils, or on spools or reels shall be packaged to afford protection against corrosion and damage from the supplier to the initial destination. The supplier may use his commercial practice providing it fulfills the above requirements.

5.2 Packing.

5.2.1 Level A or B.

5.2.1.1 Coils, (nominal wire diameter 0.204 through 0.080). Wire coils packaged as specified 5.1.1.1 and 5.1.2.1 shall require no packing.

5.2.1.2 Spools or reels (nominal wire diameters 0.079 through 0.020). Wire spools or reels packaged as specified in 5.1.1.1.3 and 5.1.2.1.3 shall require no packing.

5.2.2 Level C. The wire shall be packed to insure carrier acceptance and safe delivery at destination, complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, each coil spool or reel shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, each coil, spool or reel shall be marked in accordance with MIL-STD-129.

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Paragraph 6.1: after (j) insert the following:
(k) Standard weight coils when required (see 5.1.1.1.2 and 5.1.2.1.2).

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MILITARY INTEREST:

Under Review, delete MU and substitute ME, delete DSA.
Under Users, delete MO and substitute MU, AT, AV, delete Air Force-85

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